Youssef Sayed

AI & Machine Learning Enthusiast

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About Me

Computer Engineering student with hands-on experience in machine learning, deep learning, and AI-driven security systems. Built and deployed a forensic-grade deepfake detection platform combining CNN feature extraction, Gram Matrix analysis, and ensemble learning. Published undergraduate research on spatial-frequency forensic pipelines. Passionate about leveraging AI to solve real-world cybersecurity challenges, with a strong foundation in algorithms, data structures, and scalable software architecture.

Education

Cairo University 2024 – Present
Bachelor of Engineering – Computer Engineering

- Cumulative Grade: Excellent
- Expected Graduation: July 2029

Skills

Languages: Python, C/C++, JavaScript, SQL
AI / ML: Generative AI, LLMs, RAG, LangChain, Deep Learning, XGBoost, Scikit-learn, CNNs
Back-End & APIs: FastAPI, Flask, Node.js, REST API Design
Databases: Vector Databases (Chroma/Pinecone), PostgreSQL, MongoDB, SQLite
Security: Rate Limiting, CSP Headers, Input Validation, Authentication, File Type Enforcement
Core Concepts: OOP, Data Structures & Algorithms, Machine Learning Pipelines, Exploratory Data Analysis
Tools: Git, GitHub, Postman, Docker, Jupyter Notebook
Soft Skills: Research & Technical Writing, Team Collaboration, Leadership, Mentorship

Projects

Hakawi (AI Cultural Heritage Platform) | *RAG, LLMs, Vector Search, Voice Cloning*

- Architected an interactive platform bringing historical figures to life using real-time, voice-enabled AI avatars, grounded in verified UNESCO data to eliminate LLM hallucinations.
- Designed a robust RAG pipeline via semantic search and integrated state-of-the-art open-source TTS models for zero-shot, ultra-low latency Egyptian dialect voice cloning.
- Implemented comprehensive AI guardrails and real-time observability using Phoenix to monitor retrieval confidence, response latency, and prevent out-of-context prompt injections.

Tracera (Deepfake Detection System) | *Python, FastAPI, Machine Learning*

- Engineered a forensic-grade image analysis platform powered by GramNet v3 to distinguish human-captured imagery from GAN and Diffusion-generated artifacts.
- Implemented an advanced ML pipeline that extracts CNN feature correlations, computes Gram Matrices, and runs eigenvalue profiling via an XGBoost ensemble layer.
- Integrated end-to-end data processing: image preprocessing, model inference, and a secure FastAPI REST API to deliver structured JSON responses for real-time deepfake classification.
- Published undergraduate research paper detailing the framework's spatial-frequency mathematical pipelines and forensic benchmarks.

Telecom Customer Churn Prediction (GCI Capstone) | *Predictive Modeling, EDA*

- Developed an end-to-end predictive model for a major telecom provider to analyze 100,000+ subscriber records and proactively identify high-risk churn candidates.
- Conducted extensive EDA and formulated actionable, tier-based retention protocols prioritizing high-value customers, aiming to maximize revenue protection and minimize intervention costs.

Spaceship Titanic Predictive System | *TensorFlow, Scikit-Learn, Streamlit*

- Built an end-to-end classification pipeline to predict alternate dimension transportation utilizing complex feature engineering and logical missing value imputation.
- Developed and evaluated multiple models (including Neural Networks), subsequently deploying the top-performing model as an interactive web application via Streamlit.

Courses

Project: Generative AI Applications with RAG and LangChain 09/2026
IBM (Coursera)
- Hands-on project building Retrieval-Augmented Generation (RAG) pipelines, integrating LangChain with vector databases to enhance LLM responses with domain-specific knowledge.

Retrieval Augmented Generation (RAG) 09/2026
DeepLearning.AI (Coursera)
- Focused on building advanced RAG pipelines, vector embeddings, and optimizing data retrieval techniques for robust AI systems.

Generative AI with Large Language Models 09/2026
Amazon Web Services (AWS) & DeepLearning.AI
- Deep dive into the generative AI lifecycle, including LLM architecture, transformer models, fine-tuning techniques (PEFT, LoRA), and deploying models for scalable inference.

Global Consumer Intelligence (GCI) Data Science 08/2026
Matsuo-Iwasawa Lab U-Tokyo
- Selected for a highly competitive, intensive global track focused on data science scalability, advanced feature engineering, and deploying end-to-end machine learning models.

Introduction to Machine Learning 02/2026
Pixels (In partnership with ITIDA & CREATIVA)
- Completed a 25-hour rigorous program covering exploratory data analysis (EDA), regression, classification models, and foundational deep learning architectures.

PostgreSQL for Everybody Specialization 09/2026
University of Michigan (Coursera)
- 4-course specialization covering relational database design, advanced SQL queries, indexing strategies, JSON processing in PostgreSQL, and NoSQL architecture with Deno.

Problem Solving & Leadership

ECPC 2026 Qualifications | *Competitive Programming (C++)* 08/2026
- Secured **Rank 23 out of 300+ teams** in the Egyptian Collegiate Programming Contest qualifications.
- Consistently solved complex algorithmic challenges under strict time constraints, leveraging Graph Theory, Dynamic Programming, and Combinatorics.

Q.E.D. (Quick Easy Done) Student Initiative | *Founder & Team Lead*
- Founded and led a student initiative providing structured academic support, demonstrating strong organizational, planning, and peer-mentorship skills.